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Step 1: Create the portfolio list

```
Welcome  newcode.py  stock_portfolio_manager.py  employee_salary.py
.venv > stock_portfolio_manager.py > ...
1  portfolio = []
2  print("Your stock portfolio has been created:", portfolio)
3
```

Output:

```
default:~/Desktop/pandas_numpy_project$ /home/kasm-user/Desktop/pandas_numpy_projec
t/.venv/bin/python /home/kasm-user/Desktop/pandas_numpy_project/.venv/stock_portfol
io_manager.py
Your stock portfolio has been created: []
```

Step 2: Add a stock

```
4  def add_stock(stock):
5      portfolio.append(stock)
6      return stock
7
8  print("Added:", add_stock("AAPL"))
9  print("Added:", add_stock("TSLA"))
10
```

Output:

```
Added: AAPL
Added: TSLA

--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 1
Enter stock symbol to add: GOOGL
Added: GOOGL
```

Step 3:Remove a stock

```
def remove_stock(stock):
    if stock in portfolio:
        portfolio.remove(stock)
        return stock
    else:
        return "Stock not found"
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 2
Enter stock symbol to remove: AAPL
AAPL
```

Step 4: Search for a stock

```
def find_stock(stock):
    if stock in portfolio:
        return portfolio.index(stock)
    else:
        return "Stock not found"
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 3
Enter stock symbol to find: TSLA
Result: 0
```

Step 5: Reverse the portfolio

```
def reverse_portfolio():
    portfolio.reverse()
    return portfolio
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 4
Reversed Portfolio: ['GOOGL', 'TSLA']
```

Step 6: Show the portfolio

```
def show_portfolio():
    print("Your current portfolio:")
    for stock in portfolio:
        print("-", stock)
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 5
Your current portfolio:
- GOOGL
- TSLA
```

Step 7: Compare stock Symbol lengths

```
def compare_stock_length(stock1, stock2):
    if len(stock1) > len(stock2):
        return stock1 + " has a longer symbol."
    elif len(stock2) > len(stock1):
        return stock2 + " has a longer symbol."
    else:
        return "Both have equal length."
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 6
Enter first stock symbol: AAPL
Enter second stock symbol: TSLA
Both have equal length.
```

Step 8: Create the main menu

```
while True:
    print("\n--- Stock Portfolio Menu ---")
    print("1. Add Stock")
    print("2. Remove Stock")
    print("3. Find Stock")
    print("4. Reverse Portfolio")
    print("5. Show Portfolio")
    print("6. Compare Stock Symbol Lengths")
    print("7. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        s = input("Enter stock symbol to add: ")
        print("Added:", add_stock(s))

    elif choice == "2":
        s = input("Enter stock symbol to remove: ")
        print(remove_stock(s))

    elif choice == "3":
        s = input("Enter stock symbol to find: ")
        print("Result:", find_stock(s))

    elif choice == "4":
        print("Reversed Portfolio:", reverse_portfolio())

    elif choice == "5":
        show_portfolio()
```

```
    elif choice == "6":
        s1 = input("Enter first stock symbol: ")
        s2 = input("Enter second stock symbol: ")
        print(compare_stock_length(s1, s2))

    elif choice == "7":
        print("Exiting Portfolio Manager...")
        break

    else:
        print("Invalid choice. Try again.")
```

Output:

```
--- Stock Portfolio Menu ---
1. Add Stock
2. Remove Stock
3. Find Stock
4. Reverse Portfolio
5. Show Portfolio
6. Compare Stock Symbol Lengths
7. Exit
Enter your choice: 7
Exiting Portfolio Manager...
```